Search for events in XENON1T associated with gravitational-waves

Riya Singh and Rafael Lang
Department of Physics and Astronomy, Purdue University

(On Behalf of XENON Collaboration)
(Dated: March 28, 2022)

Multiple gravitational waves events were observed while XENON1T was taking background data. We performed a search for events in the detector associated with gravitational waves during XENON1T science run 0 and science run 1. No event was observed in Nuclear Recoils(NR), Electronic Recoils(ER), CEvNS (2fold), and s2only channels within ±500 seconds of gravitational waves.

I. INTRODUCTION

- Give information about GW especially focusing on multi-messenger astronomy
- Give summary of past attempts and what differentiates us from other detectors (might add a table of detector versus energy-range of search)

II. XENON1T DETECTOR

A short summary of XENON1T detector

III. ANALYSIS CHANNELS

1-2 line summary of each channel and refer to the papers

1. Nuclear Recoils(NR)
2. Electronic Recoils(ER)
3. CEvNS (2fold)
4. s2only

- Table summarising energy-range, event-rate and upper limit on ER, NR, s2only channel.
- Refer to Plot of ER spectra and mention energy-range defined
- Include plot of ER-rate and our background rate - mention about sideband to check consistency
- give a table for 5 bins for 4 GWs - event-rate and upper limit

IV. CONCLUSION AND FUTURE PLANS

realtime analysis with XENONnT - SNEWS collaboration